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Understanding novel fire regimes using plant trait-based approaches: An introduction

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Fire is a natural disturbance in many biomes, but altered fire regimes are a global change threat impacting both fire-sensitive and fire-dependent ecosystems. Novel fire regimes now affect many ecosystems, and as the main fuels of wildfires, plants feature prominently in our understanding of fire as part of a changing world. This issue of the *American Journal of Botany* showcases **studies using trait-based approaches to understand fire–plant dynamics from evolutionary, ecological, organismal, physiological, fire management, and conservation perspectives**.

Plant traits—in the broad sense, including functional, species, response and effect, and emergent traits—can provide mechanistic insights across biological scales, improving our understandings of many botanical aspects of novel fire regimes, including evolutionary syndromes conferring fire tolerance to plants (Triplett and David, 2024; Bombo et al., 2025; Cruz et al., 2025; Keeley and Pausas, 2025; Schwilk et al., 2025; Turck et al., 2025), plant responses to altered frequency and intensity of fires (heathlands: Simpson et al., 2025), the role of plant flammability (Mahmud et al., 2025; Murray et al., 2025; Pacheco et al., 2025; Sultana et al., 2025), plant community assembly under novel fire regimes (Raubenheimer et al., 2025; Sano et al., 2025), restoration in the context of fire, fire as a restoration tool, and several others.

Given that altered fire regimes are global phenomena, the special issue strove for a global perspective. In particular, fire is emerging as a novel threat in many regions with historically low fire impact, making these systems promising sites for novel fire regime studies, such as the Pacific Islands (Barton et al., 2025), tropical wet forests (rainforest: Burton et al., 2025; Pacuk et al., 2025), the Chilean matorral (Ocampo-Zuleta et al., 2025), high-elevation and cloud forests, alpine and polar forests, grasslands (Charton et al., 2025; Nerlekar et al., 2025), savannas (Zanzarini et al., 2025), and bogs.

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The *American Journal of Botany* is pleased to present the special issue: "Understanding novel fire regimes using plant trait-based approaches".

REFERENCES

Barton, K. E., T. Ibanez, P. J. Bellingham, and X. Moreira. 2025. Island plant fire tolerance: Functional traits associated with novel disturbance regimes. *American Journal of Botany* 112: e16465. https://doi.org/10.1002/ajb2.16465.

Bombo, A. B., A. Fidelis, and S. Le Stradic. 2025. Modifications in fire frequency impact belowground plant components in old-growth grasslands, posing risks to their resilience. *American Journal of Botany* 112: e70108. https://doi.org/10.1002/ajb2.70108

Burton, J. E., T. D. Penman, and R. J. Peacock. 2025. Fires in rainforest: Quantifying litter bed flammability of cool temperate rainforests in eastern Australia. *American Journal of Botany* 112: e70111. https://doi.org/10.1002/ajb2.70111

Charton, K. T., J. J. Henn, M. A. Homann, C. R. Warneke, and E. I. Damschen. 2025. A functional trait perspective on responses of a restored temperate grassland to changing winter insulation and managed disturbance by fire. *American Journal of Botany* 112: e70109. https://doi.org/10.1002/ajb2.70109

Cruz, W. J. A., M. S. Machado, F. Navarro-Rosales, M. A. Carniello, M. L. F. Andrade, F. C. Oliveira, and I. Oliveras Menor. 2025. Deciphering fire tolerance of trees at the Amazonia–Cerrado transition by trait-based approach: Implications from species to communities. *American Journal of Botany* 112: e70066. https://doi.org/10.1002/ajb2.70066

Keeley, J. E., and J. G. Pausas. 2025. *Sequoia* and *Sequoiadendron*: Two paleoendemic megatrees with markedly different adaptive responses to recent high-severity fires. *American Journal of Botany* 112: e70089. https://doi.org/10.1002/ajb2.70089

Mahmud, A., N. Aktepe, and D. W. Schwilk. 2025. Moisture loss rate drives the species-specific sensitivity of shoot flammability to water status. *American Journal of Botany* 112: e70052. https://doi.org/10.1002/ajb2.70052

Murray, B. R., L. K. Hardstaff, M. L. Murray, and Z. A. Xirocostas. 2025. Toward a macroevolutionary understanding of live-leaf flammability in plant species of fire-prone forests. *American Journal of Botany* 112: e70073. https://doi.org/10.1002/ajb2.70073

Nerlekar, A. N., D. Spalink, and J. W. Veldman. 2025. Grass functional traits reflect the long history of fire and grazers in the savannas of Texas. *American Journal of Botany* 112: e70013. https://doi.org/10.1002/ajb2.70013

Ocampo-Zuleta, K., S. Paula, J. G. Pausas, L. A. Cavieres, Á. Sierra-Almeida, and S. Gómez-González. 2025. Post-fire germination and plant invasion in Mediterranean Chile. *American Journal of Botany* 112: e70057. https://doi.org/10.1002/ajb2.70057

Pacheco, A. S., H. D. Goodman, L. Hankenson, A. Ortiz, H. M. Marinace, J. J. Fisk, E. A. Bischoff, et al. 2025. Fight fire with food forests: Assessing flammability of tropical crop plant species to design fire-smart agroforestry systems. *American Journal of Botany* 112: e70075. https://doi.org/10.1002/ajb2.70075

Pacuk, D., P. van der Sleen, F. J. Sterck, and M. T. van der Sande. 2025. Toward a functional understanding of novel fire regimes in tropical forests. *American Journal of Botany* 112: e70112. https://doi.org/10.1002/ajb2.70112

Raubenheimer, S. L., L. Zheng, A. Stefanski, and P. B. Reich. 2025. Climate-change-driven shifts in C3 and C4 grass distributions and leaf traits could lead to changes in community-level flammability. *American Journal of Botany* 112: e70081. https://doi.org/10.1002/ajb2.70081

Sano, M., R. Tangney, A. Thomsen, and M. K. J. Ooi. 2025. Extreme fire severity interacts with seed traits to moderate post-fire species assemblages. *American Journal of Botany* 112: e70012. https://doi.org/10.1002/ajb2.70012

Schwilk, D. W., Md. A. Alam, N. Gill, B. R. Murray, R. H. Nolan, S. Ondei, G. L. W. Perry, et al. 2025. From plant traits to fire behavior: Scaling issues in flammability studies. *American Journal of Botany* 112: e70040. https://doi.org/10.1002/ajb2.70040

Simpson, K. J., C. M. Belcher, and S. J. Baker. 2025. Adaptive plant traits under anthropogenic burning regimes: A database for UK heath and mire plant species. *American Journal of Botany* 112: e70090. https://doi.org/10.1002/ajb2.70090

Sultana, N., C. Macinnis-Ng, S. J. Richardson, Md. A. Alam, X. Cui, S. V. Wyse, D. C. Laughlin, et al. 2025. Testing key tenets of pyro-ecophysiology: Indicators of drought response in relation to shoot flammability. *American Journal of Botany* 112: e70091. https://doi.org/10.1002/ajb2.70091

Triplett, G., and A. S. David. 2024. Stomatal distribution and post-fire recovery: Intra- and interspecific variation in plants of the pyrogenic Florida scrub. *American Journal of Botany* 111: e16284. https://doi.org/10.1002/ajb2.16284

Turck, D. F., A. M. Sparks, J. Sullivan, and D. C. Tank. 2025. Community phylogenetics of North American conifers through the lens of fire-adapted traits. *American Journal of Botany* 112: e70029. https://doi.org/10.1002/ajb2.70029

Zanzarini, V., D. R. Rossatto, and A. Fidelis. 2025. Changes in plant flammability-related traits to fire regime characteristics and biomass conditions in the Cerrado. *American Journal of Botany* 112: e70110. https://doi.org/10.1002/ajb2.70110

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